

SOFTWARE TESTING ASSIGNMENT

Comprehensive QA Documentation and Case Studies

Q1. Pick any one app you use daily (like Zomato, Instagram, or Paytm) and write a detailed test plan for its login feature, including at least these components: objective, scope, test items, features to be tested, features not to be tested, test deliverables, schedule, and responsible roles.

Test Plan: Zomato Login Feature

1. Objective: The objective of this test plan is to verify that the Zomato Login feature works correctly, securely, and allows users to access their accounts using valid credentials. The system should also display proper error messages for invalid login attempts.

2. Scope: This testing covers all functionalities related to the Login feature of the Zomato application.

In Scope:

- Login using registered mobile number
- OTP verification
- Login using email and password (if available)
- Validation of empty fields
- Invalid login attempts
- Forgot Password functionality
- Error message verification

3. Test Items: The following items will be tested:

- Login Screen
- Mobile Number Field
- OTP Verification Screen
- Login Button
- Forgot Password Option
- Error Messages

4. Features to be Tested:

- Valid login with registered mobile number
- Valid OTP verification
- Invalid OTP handling
- Empty field validation
- Forgot Password functionality
- Login session management
- Error message display

5. Features Not to be Tested:

- User Registration
- Food Search Functionality
- Restaurant Listings
- Order Placement
- Payment Gateway
- Profile Management

6. Test Deliverables:

- Test Plan Document
- Test Cases
- Test Data
- Defect Reports
- Test Execution Report
- Test Summary Report
- Test Closure Report

7. Schedule:

Activity	Duration
Requirement Analysis	Day 1
Test Planning	Day 2
Test Case Design	Day 3
Test Environment Setup	Day 4
Test Execution	Day 5-6
Defect Retesting	Day 7
Test Closure	Day 8

8. Responsible Roles:

Role	Responsibility
Test Leader	Prepare test plan, monitor testing progress, review reports
Tester	Design test cases, execute tests, report defects
Developer	Fix reported defects
Product Owner	Review requirements and approve testing results

Q2. Create a mini test strategy document for the search functionality in a movie ticket booking app (like BookMyShow), specifying your approach to test levels (unit, integration, system), test

types (functional, non-functional), and tools you would use. (Hint: Mention at least one tool for each test level or type.)

Mini Test Strategy Document – Search Functionality (BookMyShow App)

1. Objective: The objective is to verify that the movie search functionality works correctly, quickly, and accurately so users can find movies, theaters, and shows without issues.

2. Test Levels:

Test Level	Approach	Tool Used
Unit Testing	Test individual search functions and search filters.	JUnit
Integration Testing	Verify interaction between Search UI, API, and Database.	Postman
System Testing	Test the complete search functionality in the application.	Selenium

3. Test Types:

Functional Testing:

- Search movie by name.
- Search theater by location.
- Search with valid and invalid keywords.
- Verify search suggestions and filters.

Tool Used: Selenium

Non-Functional Testing:

- **Performance Testing:** Check search response time under heavy load.
Tool Used: Apache JMeter
- **Usability Testing:** Verify search bar is easy to use and understand.
Tool Used: User Feedback Sessions
- **Compatibility Testing:** Verify search works on different browsers and devices.
Tool Used: BrowserStack

4. Test Environment:

- Web Browser: Chrome, Firefox
- Mobile Devices: Android, iOS
- Stable Internet Connection
- Test Database with Movie Data

5. Entry Criteria:

- Search feature development completed.
- Test environment ready.
- Test data available.

6. Exit Criteria:

- All critical test cases passed.
- Major defects fixed and retested.
- Search results displayed correctly.

7. Deliverables:

- Test Cases
- Defect Reports
- Test Execution Report
- Test Summary Report

Q3. List and explain three key differences between a test plan and a test strategy, using examples from any food delivery or e-commerce app you know.

Difference Between Test Plan and Test Strategy

Test Plan	Test Strategy	Example (Zomato/Amazon)
1. Defines What to Test	Defines How to Test	Zomato Login Feature: Test Plan lists login, OTP verification, and forgot password. Test Strategy explains that functional, performance, and security testing will be used.
2. Project-Specific Document	Organization-Level Document	Amazon Cart Feature: Test Plan is created specifically for the shopping cart project. Test Strategy can be reused for multiple Amazon projects.
3. Prepared by Test Manager/ Test Lead	Prepared by Senior QA Manager or QA Team	For a new food ordering feature in Zomato, the Test Lead creates the Test Plan, while the QA Manager defines the overall Test Strategy.

Q4. Identify at least two risks, two assumptions, and two dependencies that could affect testing the payment feature in a shopping app (like Flipkart or Myntra).

Risks, Assumptions, and Dependencies for Testing the Payment Feature (Flipkart App)

1. Risks:

- **Risk 1: Payment Gateway Failure**

The payment gateway may be unavailable or slow, causing payment transactions to fail.

Example: User pays through UPI, but the payment gateway times out and the order is not confirmed.

- **Risk 2: Security Issues**

Sensitive payment information may be exposed if security controls are not properly implemented.

Example: Credit card details are not encrypted during payment processing.

2. Assumptions:

- **Assumption 1: Test Environment is Available**

The payment gateway test environment is available and working properly.

Example: Testers can access the UPI, Credit Card, and Net Banking test accounts.

- **Assumption 1: Stable Internet Connection**

Users and testers have a stable internet connection while making payments.

Example: Payment is expected to complete successfully without network interruptions.

3. Dependencies:

- **Dependency 1: Payment Gateway Integration**

The payment feature depends on third-party payment gateways such as PhonePe, Google Pay, or Razorpay.

Example: Testing cannot proceed if the payment gateway API is unavailable.

- **Dependency 2: Bank and UPI Services**

The payment process depends on banking systems and UPI services being operational.

Example: If the bank server is down, payment transactions cannot be completed.

Q5. Given this scenario: Your team is planning to test the 'add to cart' feature in an online grocery app. Write a short paragraph describing how you would update the test plan if a new dependency (like a third-party coupon system) is introduced during the project. (Hint: Focus on changes to scope, schedule, or deliverables.)

Updating the Test Plan for a New Dependency

If a third-party coupon system is introduced while testing the **Add to Cart** feature in an online grocery app, the test plan should be updated to include coupon-related scenarios in the testing scope. New test cases should be added to verify coupon application, discount calculation, and coupon validation. The testing schedule may need to be extended because additional integration and regression testing will be required. The test deliverables should also be updated to include new test cases, defect reports, and test results related to the coupon system. This ensures that the Add to Cart feature works correctly with the new dependency before release.